

Talia Lily Ringer (they/them)

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<https://github.com/tlringer>

ACADEMIA

University of Illinois Urbana-Champaign 2021 – Present
Assistant Professor (5 YOE)

University of Washington 2015 – 2021
Ph.D. in Computer Science (advised by Dan Grossman).
Ph.D. Thesis: [Proof Repair](#). [Defense video](#).

University of Maryland, College Park 2008 – 2012
B.S. in Mathematics and Computer Science. (advised by Larry Washington).
Honors Thesis: [An Elliptic Curve Threshold Key Establishment Scheme](#)

INDUSTRY

Industrial Research (1 YOE)
Visiting Researcher, Google (N2Formal) Summer 2022 – Winter 2022
Research Scientist Intern, Amazon (Automated Reasoning Group) Summer 2016

- Worked on machine learning tools for proof assistants at Google.
- Developed a solver-aided language for automated test input generation at Amazon.

Software Engineering (4 YOE)
Software Development Engineer II, Amazon 2015
Software Development Engineer I, Amazon 2012 – 2014
Software Development Engineer Intern, Amazon Summer 2011
Software Intern, Carr Astronautics 2010 – 2011

- Helped launch [Amazon Business](#).
- Wrote customer-facing code used company-wide at Amazon.
- Deleted more lines of code during my time as an engineer at Amazon than I added. :)

INTERNATIONAL PROGRAMS FOUNDED

I am the **founder and president** of the [Computing Connections Fellowship](#), which provides institution-independent transitional funding and research visits for computer science Ph.D. students seeking healthier environments. Our two year pilot is in programming languages.

I am also the **founder and previous chair** of [SIGPLAN-M](#), a cross-institutional long-term mentoring program for the programming languages research community. It currently reaches more than **200 mentors** and **300 mentees** across more than **44 countries**, and has been described by mentees as “**life changing**” and “**a career saver.**”

My work founding these two programs (along with my other service work) led me to receive the **2023 ACM SIGPLAN [Distinguished Service Award](#)**.

RESEARCH IMPACT

My work introduced [proof repair](#), which has since been implemented and adapted at [Amazon](#) and [NASA](#), and was the subject of a [DARPA AI Exploration](#).

RESEARCH VISION

By making it easier to use **proof assistants**, my work aims to **empower collaboration in both mathematics and software engineering at a massive scale while preserving correctness**. I take a broad view of collaboration, encapsulating both human-human and human-machine collaboration. The technologies that I build for this draw on techniques ranging from **dependent type theory** to **machine learning**, always centering the needs of **real humans**.

CURRENT ADVISING

Ph.D. Students:

- [Chris Lam](#), 2021 – Present
- **Cosmo Viola**, 2021 – Present
- [Hannah Leung](#), 2022 – Present
- **Trey Plante**, 2025 – Present
- **Eric Paul**, 2025 – Present
- **Jamie Fulford**, 2025 – Present
- **Jasper Lee**, 2025 – Present
- **Kevin Fisher**, 2025 – Present
- **Priyam Sahoo**, 2024 – Present

Masters Students:

- **Eyad Loutfi**, 2024 – Present

Undergraduates:

- **Audra Aurora Izzani** (2026 – Present)
- **Daniel Gu** (2026 – Present)
- **Sumanth Kalluru** (2026 – Present)

PREVIOUS ADVISING

Postdoctoral Researchers

- [Kiran Gopinathan](#), 2024 – 2025

Ph.D. Students:

- **Dylan Zhang**, 2022 – 2024

Visiting Ph.D. Students:

- **Jilin Hu**, 2024 – 2025

Researchers:

- **Evan Marzion**, 2024 – 2025

Research Programmers:

- **Arpan Agrawal**, 2022 – 2024

Masters Students:

- **Thomas Reichel**, 2022 – 2024
 - Thesis: [Neural Approaches to Theorem Search and Proof Repair](#)

Undergraduates:

- **Eeshan Zele**, 2022 – 2024
- **Jasper Hugunin**, 2018 – 2019 (coadvised with Dan Grossman)
 - Thesis: [Constructing Inductive-Inductive Types in Cubical Type Theory](#)
- **Max Fan**, 2022 – 2024
 - Now at Cornell for Ph.D.
- **Sankar Gopalkrishna**, 2022 – 2025
- **Taylor Blau**, 2019 – 2020 (coadvised with Dan Grossman)
 - Thesis: [Verifying Strong Eventual Consistency in \$\delta\$ -CRDTs](#)
- **Timothy Zhou**, 2021 – 2024
 - Now at UCSD for Ph.D.
- **Zory Zhang**, 2023

DISSERTATION COMMITTEES

Anastasiya Kravchuk-Kirilyuk, Harvard, 2026.
[Languages, Tools, and Interfaces for Safer Code Evolution.](#)

Bolton Bailey, UIUC, 2024.
[Formalizing Soundness Proofs of SNARKs.](#)

Emily First, UMass Amherst, 2023.
[Automating the Formal Verification of Software.](#)

HONORS & AWARDS

Aristotle Principal Investigator Sponsorship Award	2026
ESEC/FSE Distinguished Paper Award	2023
DARPA Young Faculty Award	2023
ACM SIGPLAN Distinguished Service Award	2023
Amazon Research Award	2022
PEO Scholar Award	2020
NSF GRFP	2016

GRANTS AWARDED

Formalization of Open Games in Lean. Aristotle Principal Investigator Sponsorship Award 2026. PI: Talia Ringer.

TrACE-Contrast: Faithful & Consistent Code Reasoning via Trace-Aware Contrastive Learning. Capital One ASKS, 2026. PI: Talia Ringer. Co-PI: Reyhaneh Jabbarvand (UIUC).

Automated Synthetic Theorem-proving by Reasoning in Agda with Language models

(ASTRAL). DARPA ExpMath, 2026. Primary PI: Emily Riehl (JHU).

VERSE: Verification Engineering for Real-World Software Engineers. DARPA [PROVERS](#), 2024. Primary PIs: Galois.

EPSULLM: Enabling Formal Methods Proof Synthesis Using Large Language Models. DARPA Seedling, 2024. PI: Yuriy Brun (UMass Amherst). Co-PI: **Talia Ringer** (UIUC).

Relation Learning for Proof Automation – PRICELESS. DARPA Young Faculty Award 2023. PI: **Talia Ringer** (UIUC).

PLATO: Enriched Tactic Prediction Models for Proof Synthesis & Repair. DARPA [PEARLS](#) AI Exploration. PI: **Talia Ringer** (UIUC). Co-PIs: Yuriy Brun (UMass Amherst) and Alex Sanchez-Stern (UMass Amherst).

POLYMORPH: Promotion to Optimal Languages Yielding Modular Operator-driven Replacements and Programmatic Hooks. DARPA [V-SPILLS](#). Primary PIs: Galois. Co-PIs: Northeastern, UW, UIUC, Alabama, and Syracuse.

Neurosymbolic Proof Synthesis and Repair. [Amazon Research Awards](#) 2022. PI: **Talia Ringer** (UIUC).

INVITED TALKS

Big Proof Indiana	2025
Big, Pretty Math Mechanization and Mathematical Research Workshop, Lorentz Center	2025
Bridging Neural and Symbolic Proof Automation TYPES	2024
Language Models for Formal Proof Symposium on the Science of Security (HotSoS) Keynote	2024
Proofs and Conversations UIUC Math Colloquium	2024
Concrete Problems in Proof Automation AI for Theorem Proving (AITP) , EuroProofNet Large Libraries of Proofs	2022
You and Your Environment Programming Languages Mentoring Workshop (PLMW) at POPL	2022
Proof Engineering Tools for a New Era Caltech, UCLA, UMass Amherst, Aarhus, Vermont, Illinois, Virginia, Tufts	2021

Proof Repair Across Type Equivalences
Cornell, CMU, NUS

2020 – 2021

Proof Transformation
Logic Supergroup Seminar Series

2020

Proof Engineering Tools for a New Era
Rising Stars in CS Lecture Series at UMass Amherst

2019

INVITED SEMINARS AND WORKSHOPS

Lorentz Center Workshop
Mechanization and Mathematical Research

Fall 2025

Shonan Seminar
Foundation Models and Software Engineering: Challenges and Opportunities

Spring 2024

Dagstuhl Seminar
Automated Mathematics: Integrating Proofs, Algorithms, and Data

Fall 2023

Career Path Panel
Programming Languages Mentoring Workshop (PLMW) at PLDI

Summer 2023

NeurIPS Queer in AI Workshop: Queerness and Faculty Panel
Invited Panel

Winter 2022

NeurIPS Workshop on MATH-AI
Invited Panel

Winter 2022, 2023

ICML Workshop on Human-Machine Collaboration and Teaming
Invited Panel: Human-Machine Teams for Mathematicians

Summer 2022

Coq Users and Developers Workshop
An Event for Understanding, Improving, and Extending Coq

Summer 2018, 2019

Rising Stars
An Academic Career Workshop for Women in EECS

Fall 2019

REFEREED CONFERENCE & JOURNAL PUBLICATIONS

Saketh Ram Kasibatla, Arpan Agarwal, Yuriy Brun, Sorin Lerner, Talia Ringer, Emily First.
[Cobblestone: Iterative Automation for Formal Verification.](#)
ICSE 2026.

Cosmo Viola, Max Fan, Talia Ringer.
[Proof Repair across Quotient Type Equivalences](#)
OOPSLA 2025.

Borhane Blili-Hamelin, Christopher Graziul, Leif Hancox-Li, Hananel Hazan, El-Mahdi El-Mhamdi, Avijit Ghosh, Katherine Heller, Jacob Metcalf, Fabricio Murai, Eryk Salvaggio, Andrew Smart, Todd Snider, Mariame Tighanimine, Talia Ringer*, Margaret Mitchell*, Shiri Dori-Hacohen*.

[Stop Treating ‘AGI’ as the North-Star Goal of AI Research.](#)

ICML 2025 (Position Track).

Alex Sanchez-Stern, Abhishek Varghese, Zhanna Kaufman, Dylan Zhang, Talia Ringer, Yuriy Brun.

[QEDCartographer: Automating Formal Verification Using Reward-Free Reinforcement Learning.](#)

ICSE 2025.

Audrey Seo*, Chris Lam*, Dan Grossman, Talia Ringer.

[Correctly Compiling Proofs About Programs Without Proving Compilers Correct.](#)

ITP 2024.

Dylan Zhang, Curt Tigges, Zory Zhang, Stella Biderman, Maxim Raginsky, Talia Ringer.

[Transformer-Based Models Are Not Yet Perfect At Learning to Emulate Structural Recursion.](#)

TMLR 2024. [Short version.](#)

Emily First, Markus Rabe, Talia Ringer, Yuriy Brun.

[Baldur: Whole-Proof Generation and Repair with Large Language Models.](#)

ESEC/FSE 2023. **Distinguished Paper Award.**

Tom Reichel, R. Wesley Henderson, Andrew Touchet, Andrew Gardner*, Talia Ringer*.

[Proof Repair Infrastructure for Supervised Models: Building a Large Proof Repair Dataset.](#)

ITP 2023.

Alex Sanchez-Stern*, Emily First*, Timothy Zhou, Zhanna Kaufman, Yuriy Brun, Talia Ringer.

[Passport: Improving Automated Formal Verification Using Identifiers.](#)

TOPLAS Volume 45, Issue 2: No. 12, pp 1-30.

[Presented at PLDI 2023.](#) [Tool repository.](#)

Arpan Agrawal, Emily First, Zhanna Kaufman, Tom Reichel, Shizhuo Zhang, Timothy Zhou, Alex Sanchez-Stern, Talia Ringer, Yuriy Brun.

[Proofster: Automated Formal Verification.](#)

ICSE Demo 2023. [Demo video,](#) [tool website.](#)

Emily Ruppel*, Sihang Liu*, Elba Garza, Sukyoung Ryu, Alexandra Silva, Talia Ringer.

[Long-Term Mentoring for Computer Science Researchers.](#)

Communications of the ACM (CACM): Volume 66: No. 5, pp 33-35. May 2023.

Talia Ringer, RanDair Porter, Nathaniel Yazdani, John Leo, Dan Grossman.

[Proof Repair Across Type Equivalences.](#)

PLDI 2021. [Talk video,](#) PUMPKIN Pi [tool repository.](#)

Talia Ringer, Alex Sanchez-Stern, Dan Grossman, Sorin Lerner.

[REPLICA: REPL Instrumentation for Coq Analysis.](#)

CPP 2020. [Talk video.](#)

Talia Ringer, Karl Palmskog, Ilya Sergey, Milos Gligoric, Zachary Tatlock.
[QED at Large: A Survey of Engineering of Formally Verified Software.](#)
Foundations and Trends® in Programming Languages: Vol. 5: No. 2-3, pp 102-281. 2019.
[Project website.](#)

Talia Ringer, Nathaniel Yazdani, John Leo, Dan Grossman.
[Ornaments for Proof Reuse in Coq.](#)
ITP 2019. [Talk video](#), DEVOID [tool repository](#).

Talia Ringer, Nathaniel Yazdani, John Leo, Dan Grossman.
[Adapting Proof Automation to Adapt Proofs.](#)
CPP 2018. [Talk video](#), PUMPKIN PATCH [tool repository](#).

Talia Ringer, Dan Grossman, Daniel Schwartz-Narbonne, Serdar Tasiran.
[A Solver-Aided Language for Test Input Generation.](#)
OOPSLA 2017. [Talk video](#).

Talia Ringer, Dan Grossman, Franziska Roesner.
[AUDACIOUS: User-Driven Access Control with Unmodified Operating Systems.](#)
CCS 2016. [Talk video](#).

INVITED JOURNAL COMMENTARY

Talia Ringer.
[Mathematicians put AI model AlphaProof to the test](#) (News & Views).
Nature (Nov 2025).

Talia Ringer.
[Proofs and Conversations](#) (Early Career).
Notices of the American Mathematical Society 71, 9 (Oct 2024).

WORKSHOP PUBLICATIONS

Tom Reichel, Talia Ringer.
[ProofDB: A Prototype Natural Language Coq Search Engine.](#)
AITP 2024.

Dylan Zhang, Emily First, Talia Ringer.
[Getting More out of Large Language Models for Proofs.](#)
AITP 2023.

Hannah Leung, Talia Ringer, Christopher Fletcher.
[Towards Formally Verified Path ORAM in Coq.](#)
CoqPL 2023.

Seth Poulsen, Matthew West, Talia Ringer.
[Autogenerating Natural Language Proofs for Proof Education.](#)

The Coq Workshop 2022.

PUBLICATION DRAFTS

E.G. Plante III, Eric Paul, Cosmo Viola, G.A. Kavvos, and Talia Ringer.

[Proof Repair and \(Co\)algebra Equivalences.](#)

Under Submission.

Jilin Hu, Jianyu Zhang, Yongwang Zhao*, Talia Ringer*.

[HybridProver: Augmenting Theorem Proving with LLM-Driven Proof Synthesis and Refinement.](#)

Under Submission.

LEADERSHIP, DIVERSITY, & SERVICE

Service is a core part of my career since it is **so much bigger than my own research**. My service was recognized with the **2023 ACM SIGPLAN [Distinguished Service Award](#)**.

International Programs Founded:

- [Computing Connections Fellowship](#), Founder & President 2022 – 2025
- [SIGPLAN-M](#) (long-term mentoring), Founder & Previous Chair 2021 – Present

Conferences & Workshops:

- PLDI DEI Chair 2027
- Beyond Bayes Workshop Co-Chair 2022
- Coq Workshop Co-Chair 2022
- SPLASH Hybridization Committee ([first major hybrid conference in my field](#)) 2021
- ICFP Programming Languages Mentoring Workshop (PLMW) Co-Chair 2020
- ICFP Mentoring Chair 2020
- [POPLmark 15 Year Retrospective Panel](#) Lead Organizer 2020
- Shut Down PL Co-Organizer (anti-racist workshop) 2020
- DeepSpec Summer School Student Talks Organizer 2017

Departmental & University Service:

- Illinois Rhythm Syndicate Faculty Advisor 2026 – Present
- Disability Justice Panel 2023
- Grainger [IDEA Institute](#) Core Faculty Member 2022 – 2024
- UW CSE Academic Jobs Panel 2022
- Illinois CS BPC Committee 2022 – 2023
- Illinois Mental Health Ambassador 2021 – Present
- Illinois [CS CARES](#) Committee 2021 – Present
- UW CSE Care Committee Founder & Organizer (support network) 2019 – 2021
- UW CSE Graduate Admissions Committee 2018

National & International Service:

- ACM SIGPLAN Distinguished Service Award Committee 2025 – Present
- BIRS Scientific Advisory Committee 2023 – Present
- BIRS Equity, Diversity, and Inclusion Advisory Committee 2023 – Present
- NASEM [AI to Assist Mathematical Reasoning Workshop](#) Planning Committee 2023
 - [Resources Document](#)

- [C3E Workshop](#) Planning Committee

2023

Reviewing & Program Committees (PCs):

- **Panels:** NSF (2023, 2024).
- **Conference PCs:** CPP (2026), POPL (2024), ICFP (2023), ITP (2022, 2026), PLDI (2022, 2026), CAV (2021).
- **Other Conference Reviewing:** POPL (2026), ACL (2026).
- **Journal Reviewing:** Nature (2023, 2025), JAR (2022), Mathematical Structures in Computer Science (2020).
- **Workshop PCs:** TYPES (2022), AIPLANS (2021), HATRA (2020), CoqPL/RocqPL (2019, 2026).
- **Artifact Evaluation Committees:** CAV (2019), POPL (2018, 2019).

Mentoring:

- Mentor for the LEAP Alliance. 2024 – Present
- Mentor for [SIGPLAN-M](#), the program that I founded and chair. 2021 – Present
- Mentor for undergraduate and graduate women in CS at UW. 2015 – 2020
- Mentor for LGBT students through the UW Queer Mentoring Program. 2016 – 2019
- Technical and career mentor for software engineers at Amazon. 2012 – 2015

Community:

- Creator and administrator of the Neurodivergent at UIUC Slack (2022 – Present).
- Creator and administrator of the Neurodivergent in CS Slack (2021 – Present).
- Creator and administrator of the Midwest PL Slack (2021 – Present).
- Packer and food delivery driver for food banks in Seattle (2020 – 2021).
- ESL tutor & friendly visitor for an elderly refugee with JFS (2017 – 2021).
- Author of a [blog interview series](#) about LGBT CS researchers (2016 – 2018).

TEACHING

CS 498 SF: Software Foundations	<i>Fall 2026</i>
CS 598 TLR: Build Your Own Proof Assistant	<i>Spring 2026</i>
CS 576 TLR: Formalized Mathematics	<i>Fall 2025</i>
CS 576 TLR: Dependent Type Theory	<i>Spring 2024</i>
CS 421: Programming Languages and Compilers	<i>Fall 2023</i>
CS 598 TLR: Proof Automation	<i>Spring 2022, Fall 2022</i>

MEDIA

[AI Through the Lens of Journalism](#)

Live radio interview for The 21st Show.

[AI-Powered Proof Generator Helps Debug Software](#)

IEEE Spectrum article about our award-winning work on Baldur.

[Talía Ringer receives DARPA Young Faculty Award](#)

Article about receiving the DARPA Young Faculty Award.

[Ringer Awarded for Mentoring and Community Work](#)

Article about receiving the SIGPLAN Distinguished Service Award.

[Formal Verification and Deep Learning](#)

Podcast interview for The Gradient about my work as it intersects with machine learning.

[Ringer Seeks to Expand Upon the Impact of Being a Mentor](#)

Article about my international service work.

[Proof Repair](#)

Thesis Review podcast about my thesis work and how it has informed my work since.

[Tenure, Sexism, and ADHD](#)

Type Theory Forall podcast about my work and my experiences.

[How Will Proof Engineering Affect the Future of Software Development?](#)

A podcast interview about my work and future vision. From DevDiscuss Season 6, Episode 4.

[Proof Repair & Code Generation](#)

A Galois blog post by Valentin Robert about using my tools for industrial applications.

[Proof Engineering for the People](#)

A podcast interview about my work and future vision. From Building Better Systems.

[AMA on Mentoring](#)

Invited Ask Me Anything (AMA) session at ICFP 2021 about SIGPLAN-M.

[GAP Interview](#)

Interview about the academic job search.

JUST FOR FUN

I have an **amazing daughter** named Sarah Rose El-Kebir. I love her more than I could have ever imagined, and she takes up most of my time outside of work these days.

Before Sarah, I used to spend a lot of time playing **judo**. I was fairly new to it, but hooked.

Before that, I used to compete for **Club Northwest**, a top **distance running** club in Seattle. I served on the board of Club Northwest from 2015 to 2016. I ran **NCAA Division I Cross-Country** in 2009. And before even that, I was a **competitive swimmer** for ten years.

I also enjoy **roasting coffee**, **solving logic and number puzzles**, **writing poetry**, **singing**, **studying languages**, **making bagels**, **making chocolate truffles**, **windsurfing**, **playing Dance Dance Revolution and Pokemon**, **foraging edible mushrooms**, and **composing piano music**. I own a **IIDX machine** and a **Jubeat machine**.